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AQA · LEVEL 1 · MATHEMATICS

AQA Functional Skills Mathematics Level 1

March 2017 · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 45 / 45 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: AQA-4367-QP-MAR17

100% Built from Empty Question Paper + Official Mark Scheme

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HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	*Master Functional Skills (Model Exemplar)*
Centre Name:	*Master Functional Skills (MFS01)*
Lead Tutor:	*Abdul · MasterFunctionalSkills.co.uk*
Edition:	*2026 Clean Empty-QP Rebuild (Zero Collisions Verified)*

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- Stepped MFS Multi-Colour Working & Method Tips
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Level 1

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Instructions

- This copy of the Data Book is for use in the examination. It should not be given to students in advance.

Advice

- This book will not be collected in for marking. Ensure that all working that you wish to have marked is written in the space provided in the question/answer book.

4367

QAN 500/8703/4

IB/M/Mar17/E3

Data Sheet for Tenpin bowling



The game

A game has ten **frames** for each player.

In each frame, a player starts with ten pins to knock over.

A player bowls a ball one **or** two times in a frame.

If they knock over all ten pins with their first ball, their frame is complete.
This is called a **strike**.

If there are pins still standing, they bowl a second ball at the remaining pins.
If they knock them all over, this is called a **spare**.

Scoring

These examples show how to work out a player's score.

Example 1 – Sophie's first two frames

Frame 1

1st ball → 6 pins 2nd ball → 2 pins

Frame 2

1st ball → 5 pins 2nd ball → 4 pins

Here is the scoreboard after Frame 2

1		2	
6	2	5	4
8		17	
↑ 6 + 2		↑ 8 + 5 + 4	

After Frame 1, Sophie's total score is 8
After Frame 2, Sophie's total score is 17

Example 2 – Mike's first two frames

Frame 1

1st ball → 3 pins 2nd ball → 7 pins

spare

Frame 2

1st ball → 6 pins 2nd ball → 2 pins

Here is the scoreboard after Frame 1

The symbol for a spare is ⁄

1	
3	⁄

The total score after the spare cannot be worked out until Mike bowls his next ball.

A spare scores 10 + the number of pins knocked over with the player's next ball

Here is the scoreboard after Frame 2

1		2	
3	⁄	6	2
16		24	
↑ 10 + 6		↑ 16 + 6 + 2	

After Frame 1, Mike's total score is 16

After Frame 2, Mike's total score is 24

Example 3 – Fatima's first two frames

Frame 1

1st ball → 10 pins
strike

Frame 2

1st ball → 5 pins 2nd ball → 4 pins

Here is the scoreboard after Frame 1

The symbol for a strike is **X**

1
X

The total score after the strike cannot be worked out until Fatima bowls her next **two** balls.

A strike scores 10 + the number of pins knocked over with the player's next **two** balls

Here is the scoreboard after Frame 2

1		2	
X		5	4
19		28	
↑		↑	
10 + 5 + 4		19 + 5 + 4	

After Frame 1, Fatima's total score is 19
After Frame 2, Fatima's total score is 28

The scoreboard

Here is the scoreboard for a complete game for Mike.

1		2		3		4		5		6		7		8		9		10	
3	/	6	2	2	/	5	4	X		6	/	4	3	7	2	X		6	3
16		24		39		48		68		82		89		98		117		**126**	

Mike's final score is 126

Data Sheet for Electricity

Electrical items use different amounts of electricity.

The electricity used depends on

the power of the item

the amount of time the item is switched on.

Power is measured in kilowatts (kW) or watts (W).

1 kW = 1000 W

Using electricity

The number of units of electricity an item uses is worked out using

Number of units = Power in kW × Time in hours

Example

A 3000 W electric fire is switched on for $2\frac{1}{2}$ hours.

Power = 3000 ÷ 1000

= 3 kW

Time = 2.5 hours

Number of units = 3 × 2.5

= 7.5

The fire uses 7.5 units of electricity.

Cost of electricity used

Work out Number of units of electricity used × Cost per unit

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Answer **all** questions in the spaces provided.

1 Tenpin bowling

There is a **data sheet** for Tenpin bowling.

Sue, Mal and their two children are going to play a game of tenpin bowling.

1 (a) Here are the ticket prices.

Q1(a) Worked Solution:
21.16 and 21.99 and No
18.16 and No
[A2] [A1] [A1] [SC2]

Adult	£5.79
Child (under 16)	£4.79
Family (2 adults and 2 children under 16)	£18.99

Both children are under 16

Sue says,

"It is **more than** £3 cheaper to buy a family ticket than four separate tickets."

Is she correct?
You **must** show your working.

[4 marks]

Do not write outside the box

1 (b) Sue has completed two frames of the game.

1		2	
3	5	2	7

Complete the scoreboard for the first two frames.

[2 marks]

1 (c) Later, Mal has one more ball to bowl.

1		2		3		4		5		6		7		8		9		10	
6	3	8	/	6	2	5	3	7	2	1	6	5	4	8	1	**X**		3	
9		25		33		41		50		57		66		75					

Mal knocks over 5 pins with his last ball.

He says,

"My final score is **more than** 100"

Is he correct?
You **must** show your working.

[4 marks]

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I am the captain of a tenpin bowling team.

Anna

Anna has to choose a new player for her bowling team.
She can choose either Jamil or Tom.

Jamil and Tom have each played five games against Anna.
Here are their final scores and results.

Jamil	
Final Score	**Result**
145	Lost
138	Lost
204	Won
186	Won
172	Won

Tom	
Final Score	**Result**
192	Won
165	Lost
144	Lost
210	Won
184	Lost

Do not write outside the box

1 (d)



I have a greater chance of winning against Anna than Tom does.

Jamil

How do the results support Jamil's statement?

[1 mark]

1 (e)



On average, my final scores were higher than Jamil's.

Tom

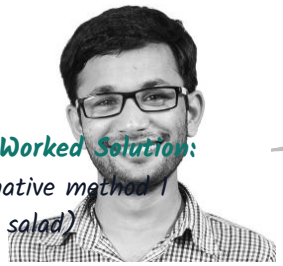
Show that Tom is correct.

[3 marks]

14

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2 Sandwiches



I make and sell sandwiches in my shop.

Q2(a) Worked Solution:
Alternative method 1
(with salad)

Amir

Amir makes sandwiches with these fillings.

Cheese Ham Tuna Egg

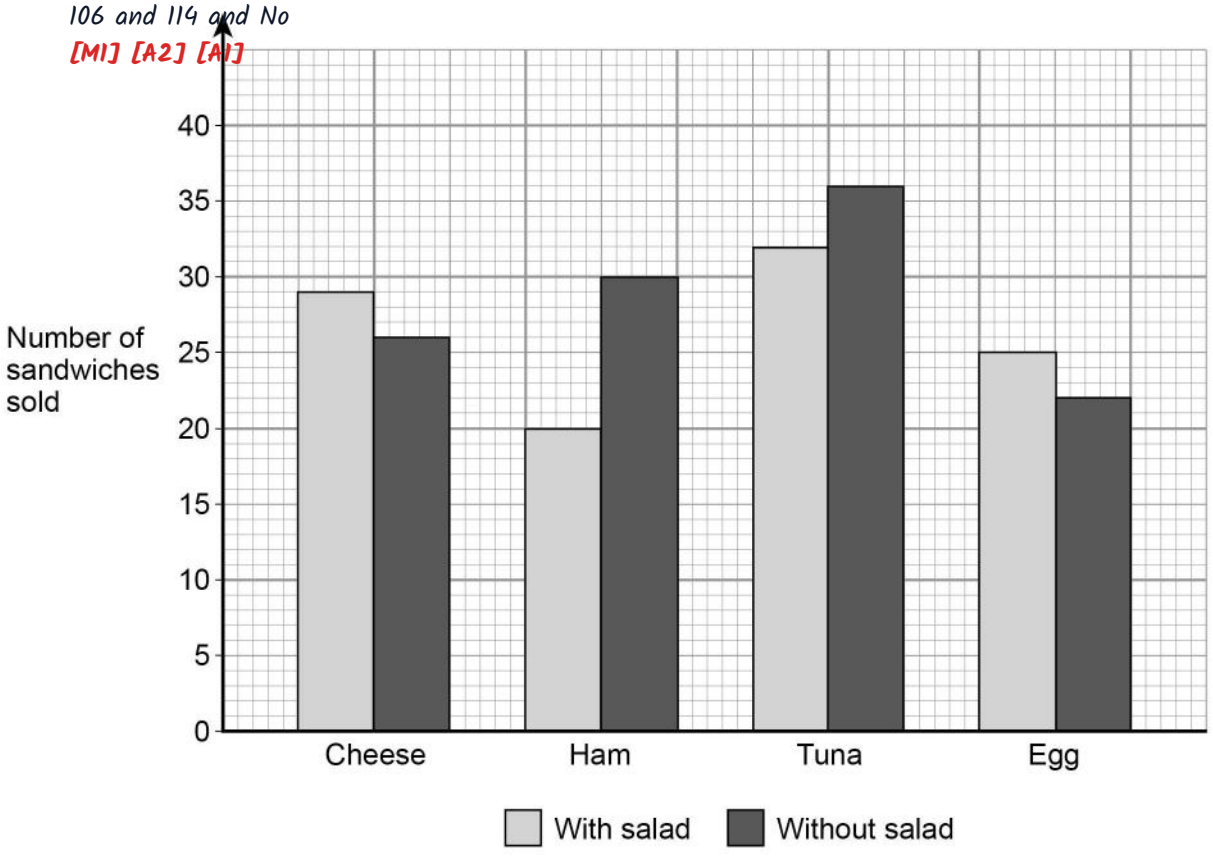
29 + 20 + 32 + 25 or 106
(without salad)

Sandwiches can be **with salad** or **without salad**.

Each week, Amir makes a chart to show the number of sandwiches sold.

Here is his chart for last week.

26 + 30 + 36 + 22 or 114
106 and 114 and No
[M1] [A2] [A1]



Do not write outside the box

2 (a) What was the highest selling filling last week?

Circle your answer.

[1 mark]

Cheese Ham Tuna Egg

2 (b) Amir says,

"Last week, I sold more sandwiches with salad than without salad."

Is he correct?
You **must** show your working.

[3 marks]

Question 2 continues on the next page

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Amir uses 300 grams of spread to make 25 sandwiches.

†2 (c) How many grams of spread does he use to make one sandwich?

[2 marks]

Check your answer.
Show how you have done your check.

[1 mark]

2 (d) Amir uses tubs of spread.

Each tub contains 2 kg of spread.

1 kg = 1000 g

Next week, Amir wants to make 500 sandwiches.

How many tubs will he need?
You **must** show your working.

[4 marks]

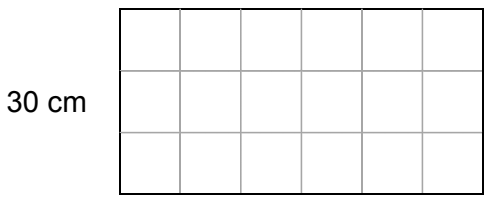
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2 (e) Amir puts the sandwiches in crates.

The crates are put on the floor space in the back of his van.

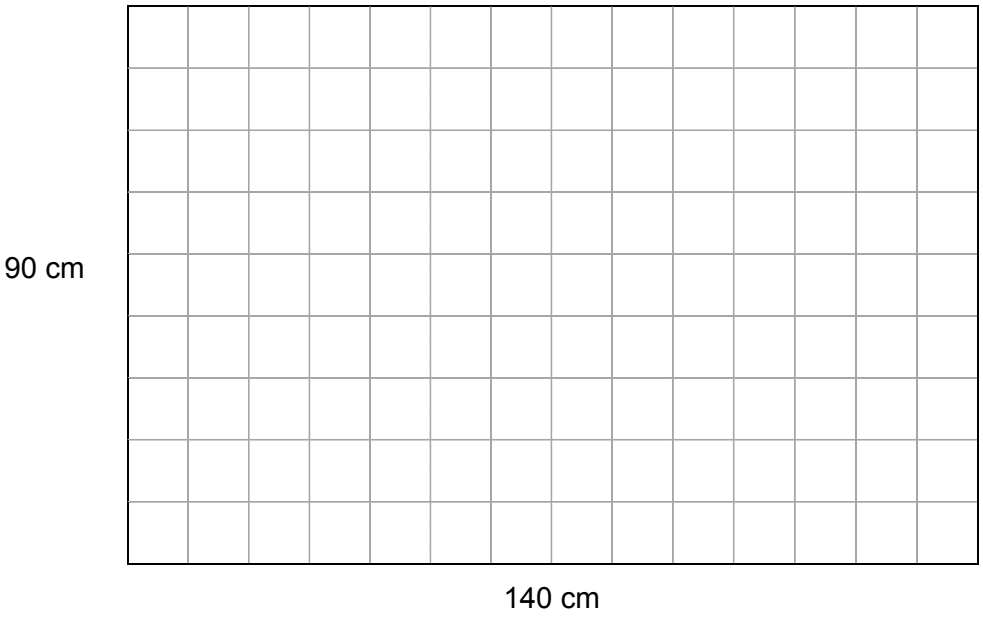
The diagrams show the base of each cuboid crate and the floor space in the van.

Base of crate



Scale ☐ represents a 10 cm by 10 cm square

Floor space of van



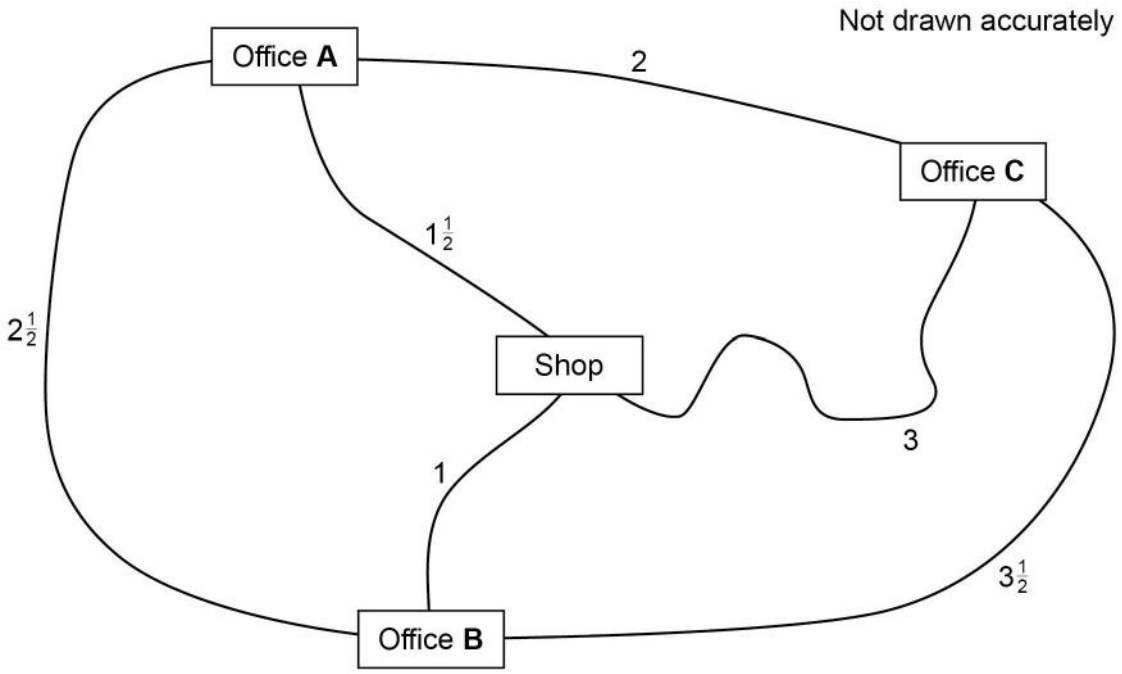
Work out the **maximum** number of crates that can fit on the floor.
You **must** show your working.

[3 marks]

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2 (f) Amir delivers sandwiches to three offices.

The diagram shows the driving distances between his shop and the offices.
The distances are in miles.



Amir drives from his shop to Office A.

He visits the other two offices.

He then drives back to his shop.

Work out the shortest possible route.
Include the route and the total distance he drives.

[3 marks]

17

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3 Hairdressing salon



I am opening a hairdressing salon.

Jenny

3 (a) Jenny is designing the salon.

Each sink will need a square space with sides of 0.5 m

Each chair will need a circular space with radius 0.5 m

She wants

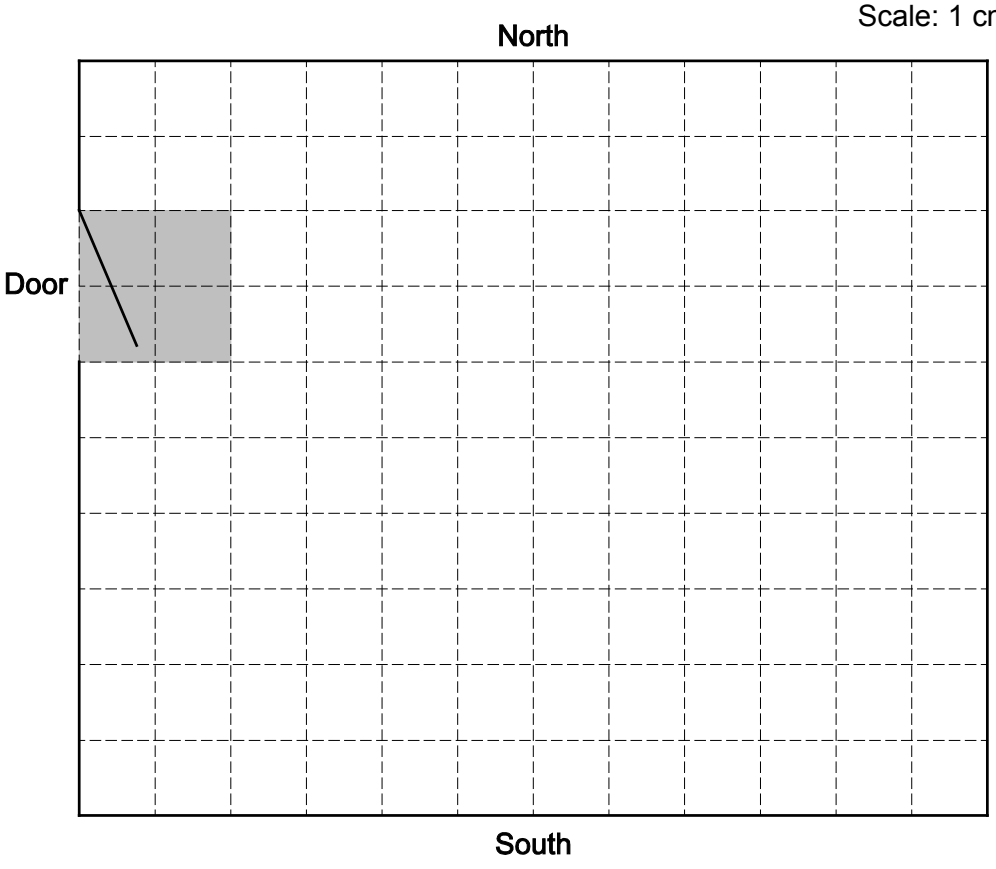
- two sinks against the North wall
- three chairs in a line in the South half of the salon
- a rectangular reception desk measuring 1 m by 0.5 m
- a rectangular waiting area measuring 2 m by 1 m

The area near the door (shaded) must be kept clear.

Show a possible design on the scale drawing below.

[5 marks]

Scale: 1 cm = 0.5 m



Question 3(a) · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

Door area kept clear and all items labelled correctly

[B1]

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3 (b) The salon opens six days each week, from Monday to Saturday.
There are two shifts each day.
Two people work each shift.

Jenny, Craig and Mia work in the salon.

Each week,

- each person works 8 shifts
- Craig does **not** work on Saturday
- each person has at least 1 full day off.

Make a possible rota for one week.

[4 marks]

Practise on this grid.

	Morning shift		**Afternoon shift**	
	Worker 1	**Worker 2**	**Worker 1**	**Worker 2**
Monday				
Tuesday				
Wednesday				
Thursday				
Friday				
Saturday				

Do not write outside the box

Put your answer on this grid.

	Morning shift		**Afternoon shift**	
	Worker 1	**Worker 2**	**Worker 1**	**Worker 2**
Monday				
Tuesday				
Wednesday				
Thursday				
Friday				
Saturday				

Question 3 continues on the next page

Do not write outside the box

3 (c) Here are the prices at the salon.

Cut and blow dry	£33
Cut and colour	£60

In the first week the salon has these bookings

30 for a cut and blow dry
12 for a cut and colour.

Each week Jenny will pay

- £704 in wages
- £300 in other costs.

Jenny says,

"My profit will be **more than** £700 in the first week."

Is she correct?
You **must** show your working.

[6 marks]

15

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4 Electricity

There is a **data sheet** for Electricity.

4 (a) Joe has an electric fire with three settings.

Q4(a) Worked Solution:

16.(..) comes from calculating the number
Allow any rounding including rounding up
Answer 270p and Yes scores M2A2
Answer 270 and 250 and Yes scores M2A2
Answer 270 with no units and Yes scores
6 × 15 (= 90) oe scores M0

Use of 3000 × 6 needs division by 1000 for
Example
3000 × 6 = 18000
18000 × 0.15 = 270 Yes M2 A0 A1ft

Setting	Power
High	3 kW
Medium	2 kW
Low	1 kW

In very cold weather he puts the fire on **High** for 6 hours each night.

Electricity costs 15p per unit.

He says,

"6 hours on High will cost **more than** £2.50"

Is he correct?
You **must** show your working.

[4 marks]

Use of 2kw or 1kw instead of 3kw clearly
Alternative method 1
[M1] [M1] [A1]

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Joe uses an ordinary 100 W light bulb in a lamp.
When the bulb wears out he replaces it with a new 100 W bulb.

The lamp is switched on for 500 hours each year.

4 (b) Show that the lamp uses 200 units of electricity in **4 years**.

[3 marks]

Joe is thinking of changing to a low energy bulb in the lamp.

A low energy bulb uses $\frac{1}{5}$ of the electricity of an ordinary bulb.

†4 (c) How many units of electricity will a low energy bulb use in **4 years**?

[2 marks]

Check your answer.
Show how you have done your check.

[1 mark]

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4 (d) Joe looks at the cost of buying and using ordinary bulbs or a low energy bulb for the next **4 years**.

The cost of buying and using ordinary bulbs will be £38

A low energy bulb

lasts for 4 years
costs £13.88

Electricity costs 15p per unit.

He says,

"Buying and using a low energy bulb is cheaper by **more than** £20"

Is he correct?
You **must** show your working.

[4 marks]

END OF QUESTIONS

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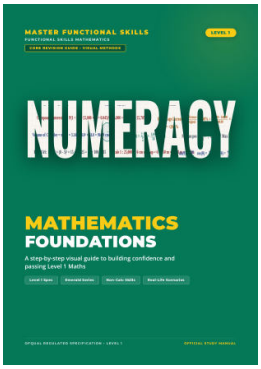
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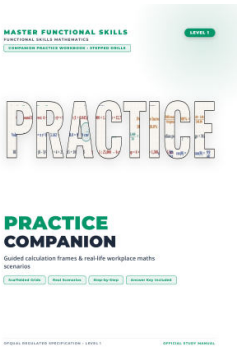


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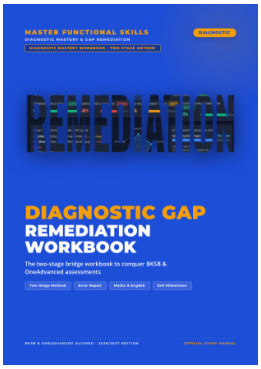


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